



Interpretation of ensemble learning to predict water quality using explainable artificial intelligence

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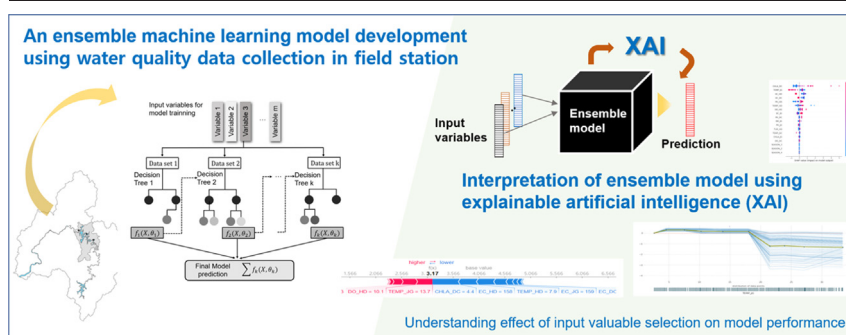
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HIGHLIGHTS

- XGBoost ensemble learning model predicts chlorophyll-*a* concentration in stream.
- XAI algorithm used to interpret ensemble learning model
- Effect of input variable selection on model performance analyzed using three indices
- Input variable selection using Shapley value provided most stable model performance.

GRAPHICAL ABSTRACT



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ABSTRACT

Algal bloom is a significant issue when managing water quality in freshwater; specifically, predicting the concentration of algae is essential to maintaining the safety of the drinking water supply system. The chlorophyll-*a* (Chl-*a*) concentration is a commonly used indicator to obtain an estimation of algal concentration. In this study, an XGBoost ensemble machine learning (ML) model was developed from eighteen input variables to predict Chl-*a* concentration. The composition and pretreatment of input variables to the model are important factors for improving model performance. Explainable artificial intelligence (XAI) is an emerging area of ML modeling that provides a reasonable interpretation of model performance. The effect of input variable selection on model performance was estimated, where the priority of input variable selection was determined using three indices: Shapley value (SHAP), feature importance (FI), and variance inflation factor (VIF). SHAP analysis is an XAI algorithm designed to compute the relative importance of input variables with consistency, providing an interpretable analysis for model prediction. The XGB models simulated with independent variables selected using three indices were evaluated with root mean square error (RMSE), RMSE-observation standard deviation ratio, and Nash-Sutcliffe efficiency. This study shows that the model exhibited the most stable performance when the priority of input variables was determined by SHAP. This implies that on-site monitoring can be designed to collect the selected input variables from the SHAP analysis to reduce the cost of overall water quality analysis. The independent variables were further analyzed using SHAP summary plot, force plot, target plot, and partial dependency plot to provide understandable interpretation on the performance of the XGB model. While XAI is still in the early stages of development, this study successfully demonstrated a good example of XAI application to improve the interpretation of machine learning model performance in predicting water quality.

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1. Introduction

Excessive increases in algae often cause a variety of harmful effects on water quality and related water supply systems, such as unfavorable taste or odor (Shen et al., 2011; Smith, 2016). Some algae, such as cyanobacteria, release substances that are toxic to human health (Cui et al., 2016; Hyun et al., 2016; Otten et al., 2015). Thus, predicting algal blooms is essential to manage water supply systems. However, measuring algal concentration is time-consuming and labor-intensive, where chlorophyll-*a* concentration (Chl-*a*) is often used as an indicator that represents the status of algal blooms in water resources (Boyer et al., 2009).

In recent years, advanced machine learning (ML) algorithms have been used to predict water quality, including Chl-*a* (Park et al., 2020a). Wu et al. (2014) used artificial neural networks (ANN) to predict daily Chl-*a* in a German lowland river where observed climate data (air temperature, daily precipitation, etc.) and water quality data (e.g., nitrate, nitrite, and total phosphorous [TP]) were used as independent input variables. Also, Huang et al. (2015) used an ANN to predict monthly Chl-*a* in a lake. Support vector machine is another ML algorithm that is used to predict water quality, such as total nitrogen, TP, and Chl-*a* (Liu and Lu, 2014; Park et al., 2015). Recently, deep learning (DL) algorithms such as long short-term memory (LSTM) have been used to predict Chl-*a* (Li et al., 2018; Shin et al., 2020; Zhou et al., 2018).

The ANN was among the first ML algorithms that were developed based on the idea of the perceptron (McCulloch and Pitts, 1943; Rosenblatt, 1958). An ANN model's structure is composed of three layers: input, hidden, and output. The hidden layer, a key structure of the algorithm, is composed of nodes, and each node calculates the output value for the next step using a non-linear function called the activation function. An increase in the number of hidden layers enables more complicated calculations or predictions from the input variables. However, the increase in the number of hidden layers is limited as it causes problems such as overfitting to the training data and vanishing gradient during the optimization of the models. DL was developed by overcoming the limitations of ANN. The DL algorithm uses an alternative activation function, such as a rectified linear unit instead of the conventional sigmoid function, which reduces the gradient vanishing problem. Also, it uses the dropout method; selective use of the nodes inside the hidden layers reduces the overfitting problem (Hinton et al., 2006; Nair and Hinton, 2010). LSTM is a popular DL algorithm based on recurrent neural networks (RNN) that overcomes the limitations of conventional RNN by selectively using information from the previous hidden layers with structures referred to as input, forget, and output gates (Greff et al., 2016). Several recent studies have used LSTM to predict water quality (Li et al., 2018; Zhou et al., 2018).

Recently, tree-based ensemble ML models have been increasingly used to predict water quality (Lu and Ma, 2020; Shin et al., 2020). Ensemble ML models are based on the theoretical idea that using a group of models composed of multiple weak learners can improve the ultimate result of the collective model, which is referred to as a strong learner (Sutton, 2005). Random forest (RF) and gradient-boosted decision tree (GBDT) are commonly used tree-based ensemble models where the decision tree models are used as weak learners. XGBoost (XGB) is among the popular and widely used GBDT models (Chen and Guestrin, 2016; Zhang et al., 2018).

An important consideration in utilizing advanced ML algorithms (e.g., ensemble ML models and DL models) for water quality prediction is that such models rely less on the physical or chemical relationships between independent input variables and target variables for the prediction, focusing instead on enabling the estimation of complicated non-linear relationships between input and target variables. However, the increased complexity of the model structure may reduce the model's interpretability. Recent studies have addressed this issue using explainable artificial intelligence (XAI) to provide understandable and reasonable explanations for model prediction results, thus increasing the confidence of model performance for practical applications (Gunning et al., 2019; Ribeiro et al., 2016; Shrikumar et al., 2016).

XAI is an emerging field of study to improve the interpretability of ML models (Adadi and Berrada, 2018; Arrieta et al., 2020). Shapley value (SHAP) analysis is a novel XAI method that provides understandable explanations of why a model determines a predictive result based on input variables (Lundberg et al., 2018; Lundberg and Lee, 2017). Partial dependency plot (PDP) is also an interpretable method used to show the marginal effect of one or two features on the predicted outcome of various ML algorithms. PDP helps understand variable importance in a given ML model and express whether a specific input variable positively or negatively affects the target value in models (Friedman, 2002; Tan et al., 2020).

Moreover, selecting input variables is a key step in developing ML models, the results of which are affected by input data in the selected input variables. Thus, the proper selection of input variables is an essential process. Tree-based ensemble ML algorithms such as RF and XGB provide a special functionality that calculates the importance of input variables against the target variable, which is referred to as feature importance (FI). FI calculates the absolute importance of each input variable on the model prediction result. However, FI cannot provide information about whether the input variable increases or decreases the model prediction value, and this is considered a limitation of FI.

However, SHAP analysis provides information about whether a given input variable influences the increase or decrease in the model prediction value. Variance inflation factor (VIF) is a statistical method that is used for input variable selection. VIF estimates the relationship between a target input variable and other input variables from linear regression. If a target input shows a close linear relationship with the other input variables, the target variable is removed from the model input data set.

The Geum River is the third-longest river in South Korea which flows for about 400 km from the middle of the Korean Peninsula to the West Sea of South Korea. It is an important water resource for drinking water, agriculture, and industrial supply in the mid-Korean region. Thus, properly managing the water quality of the Geum River is an ongoing issue, and various efforts and research are continuing for this purpose. Lim et al. (2015) used the reflection value of satellite spectral bands to predict the total nitrogen (TN) and total phosphorus (TP) in the Geum River. They first calculated the Pearson correlations between the reflection values of satellite spectral bands to select optimal sets of independent variables, and then developed multiple linear regression models to predict TN and TP. The root mean square error (RMSE) values of the models were 0.73 and 0.02 for TN and TP, respectively. Recently, Lee and Lee (2018) developed three DL models (multilayer perceptron [MLP], RNN, LSTM) and an ordinary least square (OLS) regression model for predicting the Chl-*a* concentration in 16 field monitoring sites in South Korea, including two sites (Sejong and Gongju) in the Geum River. They found that LSTM exhibits the best prediction performance, and the average RMSE values of OLS, MLP, RNN, and LSTM for the 16 sites were 17.75, 16.42, 16.13, and 16.09, respectively. The RMSE values of the LSTM model for the Sejong and Gongju sites were 30.83 and 31.92, respectively. Further, Lee et al. (2018) developed a simple ANN model using several global climate indices as independent variables, to predict rainfall during May and June in the Geum River watershed.

In this study, we first collect 16 input variables of water quality parameters for monitoring the water quality in three water bodies in South Korea. Second, an ensemble ML model is developed to predict the Chl-*a* concentrations, which represent algal concentrations. Third, the importance of input variables for model prediction is computed using three indices: SHAP, FI, and VIF. Fourth, the model input variables are determined from the three indices, and the effects of selecting input variables on model performance are compared. Finally, XAI methods are used to understand how the model determines prediction values and to estimate how information is used to improve model performance. We believe the model comparison with indices could provide the best down-selection strategy for input variables required for collecting field data, while maintaining the prediction accuracy.

2. Methods

2.1. Data sources

Water quality data measured at three parallelly located field monitoring stations upstream of Geum River, South Korea between October 2017 and March 2021 were used as input variables to develop an XGB model. The three monitoring stations were Jangge (JG), Daechung (DC), and Hyundo (HD), which were located in the mid-upstream of the Geum River (Fig. 1). Water flows from upstream JG to downstream HD station.

The data were obtained from the Real Time Water Information System of Korea (RTWIS, n.d.), which is managed by the National Institute of Environmental Research of Korea. The input variables were temperature (TEMP), pH, electrical conductivity (EC), dissolved oxygen concentration (DO), and Chl-*a*, which were measured at the three monitoring stations (Table 1). Moreover, turbidity was measured at HD station.

2.2. Data preparation

2.2.1. Dummy variables

Water quality is affected by various environmental factors, including the climate. As there are four distinct seasons with different rainfall patterns as well as air temperatures in South Korea, seasonality was considered in the development of the model by using dummy variables. The dummy variables were included using one-hot encoding, where the value was one for the designated season and zero for the other variables (Table 2). As three variables represented seasonality, no dummy variable was designated for spring.

2.2.2. Data treatment for model development

The input variables observed from October 2017 to March 2020 and from April 2020 to March 2021 were used for training and testing the XGB model, respectively. The ratio of data used for training and testing the model was 0.71:0.29, where observations during an entire year were included for both training and testing data.

The field observations used for the input variables included missing values where there were no observations. The missing values for the training data ranged approximately from 1.1% to 5.5% for each input variable. The missing values in the training data were replaced with *k*-nearest neighbor (KNN) imputation. No imputation was processed for the test data; the test data with missing values were excluded from the model prediction. Scikit-learn, a Python open-source ML library, was used for KNN imputation (Pedregosa et al., 2011).

2.3. Model development

2.3.1. Model selection

GBDT, a tree-based ensemble model, was used to develop the Chl-*a* prediction model (Fig. 2). GBDT is one of the popular ensemble models, along with RF. Both models were used as regression models and had been used for predicting water quality with good performance (Shin et al., 2020; Uddameri et al., 2020; Wang et al., 2021). The model process of the GBDT is often comparable to that of RF, which uses the bagging method for selecting input variables that are used for generating a decision tree for predicting the target value (Hollister et al., 2016; Shin et al., 2020). The average of individual decision tree predictions is used as the final

prediction result of the model. GBDT assigns a higher weight to data with higher residual errors; thus, improving model performance in the next step, where each tree is called a weak learner, which is an important difference from RF (Friedman, 2001; Ke et al., 2017). As GBDT includes a residual error in the previous stage for generating the next decision tree, the model performance improves sequentially.

The GBDT is optimized to minimize a defined objective function (Eq. (1)). The loss function (*l*) calculates the difference between the observation (*y_i*) and model prediction (*ŷ_i*) in each decision tree for all data samples (*n*). The regularization function (*Ω*) is used to penalize the complexity of the model, where *f_k* represents the function of the *k*-decision tree (Chen and Guestrin, 2016; Park et al., 2020b; Shin et al., 2020; Zhang et al., 2018). The model process for the final prediction of GBDT is shown in Fig. 2, where *X* is an input variable and *θ_k* is an independent and identically distributed random vector (Zhang et al., 2018).

$$J = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (1)$$

In this study, the XGB regressor algorithm was used to develop a model for predicting Chl-*a*. XGB was initially developed by Chen and Guestrin and is one of the widely used GBDT algorithms, where the Scikit-learn and XGB libraries were used in the model development process (Chen and Guestrin, 2016; Pedregosa et al., 2011; XGBoost, n.d.; Zhang et al., 2018).

2.3.2. Model optimization

Eighteen input variables were used as independent variables for model development to predict Chl-*a* at the HD station (Table 3). The hyperparameters of the XGB model were optimized by implementing a grid search method with ten-fold cross validation using the Scikit-learn grid search library (Pedregosa et al., 2011).

2.3.3. Model evaluation

The XGB model performance for Chl-*a* prediction was evaluated using three evaluation indices: RMSE, RMSE-observation standard deviation ratio (RSR), and Nash–Sutcliffe Efficiency (NSE) (Eqs. (2)–(4)). A smaller RMSE represented a better model prediction performance. The value of RSR ranged from 0 to 1. A smaller RSR represented a better model prediction performance, where the model was considered satisfactory when RSR < 0.70 (Bennett et al., 2013; Moriasi et al., 2007). The value of NSE ranged from -∞ to 1, where the model was considered to show a better fit when NSE approached 1.

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (M_{i,obs} - M_{i,model})^2}{n}}, \quad (2)$$

$$RSR = \frac{\sqrt{\frac{\sum_{i=1}^n (M_{i,obs} - M_{i,model})^2}{n}}}{\sqrt{\frac{\sum_{i=1}^n (M_{i,obs} - \bar{M}_{i,obs})^2}{n}}}, \quad (3)$$

Table 1
Observed input variables of water quality.

Site	Input variable
Jangge (JG)	TEMP, pH, EC, DO, Chl- <i>a</i>
Daechung (DC)	TEMP, pH, EC, DO, Chl- <i>a</i>
Hyundo (HD)	TEMP, pH, EC, DO, Chl- <i>a</i> , TUR

Table 2
Dummy variables.

Season	Spring	Summer	Fall	Winter
Period	March to May	June to August	September to November	December to February
Variable name	–	SEASON_2	SEASON_3	SEASON_4

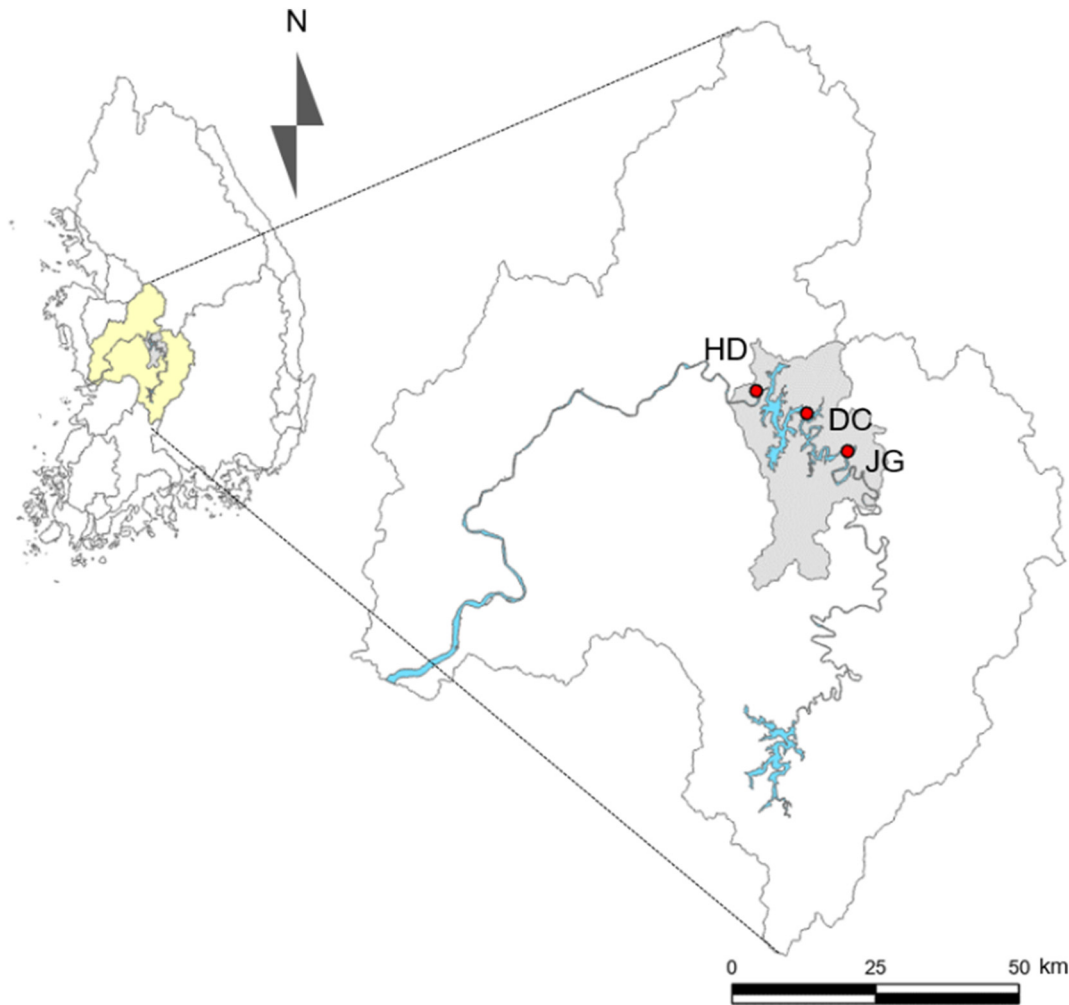


Fig. 1. Site locations of this study.

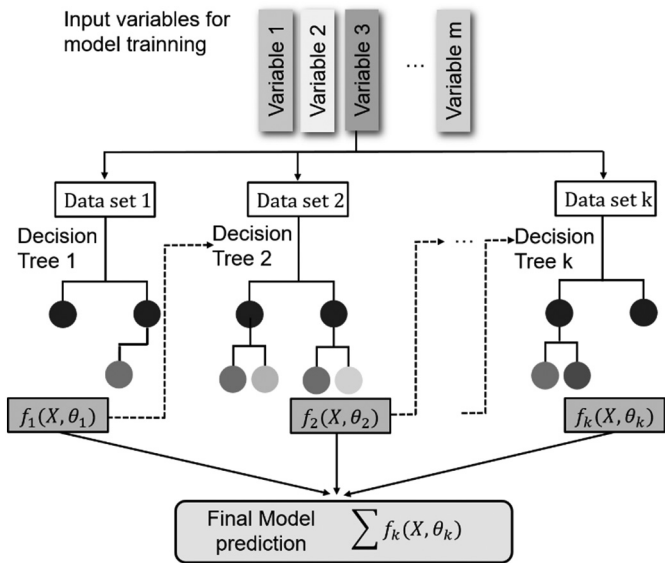


Fig. 2. Schematics of the GBDT data processing.

$$NSE = 1 - \frac{\sum_{i=1}^n (M_{i,obs} - M_{i,model})^2}{\sum_{i=1}^n (M_{i,obs} - \bar{M}_{i,obs})^2} \quad (4)$$

where $M_{i,model}$: model-predicted value;
 $M_{i,obs}$: observed value;
 $\bar{M}_{i,obs}$: mean of observed values.

2.4. XAI and input variable selection

2.4.1. SHAP

Given that insufficient interpretability is one of the important limitations to the use of ML algorithms, XAI can provide understandable explanations for how a ML model determines the model performance. SHAP

Table 3
Input variables used for model development.

Independent variables	Dependent variables
DO_HD, EC_HD, PH_HD, TEMP_HD, TUR_HD, CHLA_DC, DO_DC, EC_DC, PH_DC, TEMP_DC, CHLA_JG, DO_JG, EC_JG, PH_JG, TEMP_JG, SEASON_2, SEASON_3, SEASON_4	CHLA_HD

Note: _HD, _DC, and _JG represent monitoring station names as Hyundo, Daechung, and Jange, respectively.

analysis is one of the recently developed XAI algorithms (Lundberg et al., 2018; Lundberg and Lee, 2017). SHAP analysis is based on game theory, in which a combination of different input variables is constructed to clarify the importance of a target input variable by comparing the mean change in model output according to the presence or absence of that input variable (Lundberg et al., 2018; Lundberg and Lee, 2017). The SHAP of each input variable is the weighted mean of the marginal contribution of that input variable (Eq. (5)) (Lundberg et al., 2018; Lundberg and Lee, 2017).

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(F-|S|-1)!}{|F|!} [g_{S \cup \{i\}}(x_{S \cup \{i\}}) - g_S(x_S)] \quad (5)$$

where

ϕ_i : SHAP of the i th input variable;

F : the set of all input variables;

S : all subsets of input variables without the i th input variable.

In Eq. (5), x_S represents the values of the input variables in S without the i th input variable, whereas $x_{S \cup \{i\}}$ represents the data set that includes the i th input variable, and $g_S(x_S)$ represents the prediction based on input x_S . In this study, SHAP analysis was used to understand the effects of input variables on model performance.

2.4.2. Partial dependency plot (PDP)

PDP was designed to interpret the relationship between the input and predicted values of learning models. It is universally applied to various model analysis methods. By using the specific variables that constituted the data, it is possible to confirm whether the relationship between the features and the predicted value (target) is linear or monotonically increasing under the learned model. Thus, by linearly transforming the numerical values of variables, the effect of changes in variable values on the model can be visually interpreted. Eq. (6) presents the formula for PDP.

$$\hat{f}_{x_c}(x_s) = E_{x_c}[\hat{f}(x_s, x_c)] = \int \hat{f}(x_s, s_c) dP(x_c), \quad (6)$$

where x_s denotes variables that the partial dependence function needs to visualize, and x_c represents the variables used by the ML model ($\hat{f}$). Generally, x_c comprises one or two variables and set S is the set of variables for which we aimed to identify the degree of partial dependence. The PDP for set S was drawn by fixing the x_c variable and calculating the average by reflecting all values for x_c in the model.

2.4.3. Selecting input variables

To predict Chl-*a*, the XGB model was developed using 18 input variables. ML predicted output values based on information obtained from input variables. Thus, the model performance was affected by the selection of input variables. Here, the effect of input variable selection on the XGB model performance was analyzed using three different indices (i.e., SHAP, FI, and VIF), which estimated the relative importance of the input variables on model performance.

FI is the relative importance of the input variables to model performance. The FI is calculated as the total impurity reduction of the prediction model using the input variable (Pedregosa et al., 2011). In this study, FI was computed using the algorithms provided by the XGB library.

The VIF is used to reduce the multicollinearity of the input variables. The VIF for an input variable is calculated from the coefficient of determination (R^2) of a linear regression model developed from all the other input variables (Eq. (7)). Thus, the VIF is calculated for all input variables independently. A higher VIF of an input variable indicates that the input variable has higher correlation with other input variables. Thus, the input

Table 4

Characteristics of input variables.

Variables		Average	Min	Max	Standard deviation
Independent variables	DO_HD (mg/L)	7.73	0.50	12.50	2.57
	EC_HD(μS/cm)	148.62	85.00	182.00	22.47
	PH_HD	7.12	6.50	7.70	0.25
	TEMP_HD(°C)	11.85	4.00	22.80	5.18
	TUR_HD(NTU)	1.43	0.10	23.70	2.87
	CHLA_DC(mg/m ³)	8.65	1.10	85.40	8.89
	DO_DC(mg/L)	9.82	4.10	14.90	1.75
	EC_DC(μS/cm)	150.68	72.00	220.00	22.68
	PH_DC	7.94	6.60	10.40	0.83
	TEMP_DC(°C)	16.88	3.20	33.70	8.14
	CHLA_JG (mg/m ³)	16.41	2.50	83.90	11.05
	DO_JG (mg/L)	10.35	4.90	15.80	1.90
	EC_JG (μS/cm)	163.81	95.00	227.00	21.54
	PH_JG	7.80	6.50	9.90	0.61
	TEMP_JG (°C)	15.57	1.60	33.40	8.49
Dependent variable	CHLA_HD (mg/m ³)	3.97	0.80	22.90	2.25

Note: HD, DC, and _JG represent the monitoring station named as Hyundo, Daechung, and Jangge, respectively.

variable with a higher VIF was excluded to reduce the multicollinearity of the model.

$$VIF = \frac{1}{1-R^2} \quad (7)$$

3. Results and discussion

3.1. Characteristics of input variables

The characteristics of all observed input variables between October 2017 and March 2021 are summarized in Table 4. The 15 independent variables (Table 4) and three dummy variables (Table 2) were used for the prediction of the dependent variable (CHLA_HD).

3.2. Model simulation

An XGB model was developed using 18 independent input variables to predict Chl-*a* at the Hyundo station (CHLA_HD). The hyperparameters of the model were determined using a grid search approach where we used a learning rate of 0.1, an n estimator of 100, and a maximum depth of 2 with the training data. The performance of the model was evaluated with the test data using three indices (Table 5).

3.3. Model simulation with selected independent variables

3.3.1. Relative importance of input variables

The values of the three indices for the relative importance of the input variables to model performance are summarized in Table 6. The SHAP_ABS, that was the sum of the absolute SHAP values of each observation, was calculated from testing data. The SHAP_ABS and FI were sorted from high to low values, while VIF was sorted from low to high. The higher values of SHAP_ABS and FI indicate that the variable has a higher effect on the model prediction. On the other hand, a higher VIF indicate that the variable has a stronger linear relationship with the other variables; thus, it can be removed from the model development to reduce the multicollinearity of model performance. The highest relative importance was observed for

Table 5

Summary of model evaluation results.

RMSE	RSR	NSE
1.872	0.630	0.603

CHLA_DC using the SHAP. However, CHLA_DC was estimated as the fourth most important input variable when FI was used. PH_DC was determined as the variable with the highest importance when FI was used. Seasonality had the least effect on model performance according to both SHAP_ABS and FI, indicating that seasonality may not be an important variable on Chl-*a* prediction. The lowest VIF was observed for TUR_HD, and highest VIF was observed for TEMP_DC. SHAP and FI suggested a different relative importance of input variables, while both indices showed that CHLA_DC, EC_HD, and PH_HD were included in the five most important input variables for model performance.

3.3.2. Comparison of the model performance with selected independent variables

The XGB model was simulated with the use of 2 to 15 selected independent variables following the order of highest SHAP and FI values, respectively, and was also simulated following the order of lowest VIF value. The performance of the XGB model was compared using three evaluation indices: RMSE, RSR, and NSE.

For model simulation starting with a small number of input variables selection from 2 to 5, the use of input variables with a higher SHAP_ABS showed a better performance compared to FI and VIF (Fig. 3).

Also, the model showed more stable performance when SHAP analysis was used for input variable selection compared to FI and VIF. The RSR of the model performance was 0.74 with the first two input variables (i.e., CHLA_DC and TEMP_JG) selected by SHAP analysis, and was then improved to 0.63 when the next variable (i.e., EC_HD) was included in the model. The RSR variation of the SHAP analysis was minimal from 0.61 to 0.65 when additional input variables from 3 to 15 were used for model simulation (Fig. 3).

On the other hand, FI and VIF analysis showed poor model performance with fewer than four input variables selected and the RSR values were 0.99 and 1.00 with the first two input variables that were determined from FI and VIF analysis, respectively. FI and VIF indices showed similar results. Between FI and VIF, a better model performance improvement was observed by increasing the number of input variables based on FI analysis compared to VIF analysis. VIF is a traditional statistical approach that is widely used for input variable selection to reduce the multicollinearity of model performance. VIF was determined from the relationship between input variables rather than the effect on model performance, whereas FI and SHAP were determined from the effect of input variables on model structure and performance. The different analytical approaches of the three indices can be considered as the main reason for variation in model performances from input variables selection. A previous study showed an example of down-selection of input variables while maintaining a stable model performance.

Table 6
Summary of indices for relative importance of input variables.

SHAP_ABS		FI		VIF	
Variable	Value	Variable	Value	Variable	Value
CHLA_DC	0.81	PH_DC	0.16	TUR_HD	2.22
TEMP_JG	0.55	EC_HD	0.13	CHLA_JG	2.73
EC_HD	0.52	DO_HD	0.12	EC_JG	2.86
EC_DC	0.36	CHLA_DC	0.12	CHLA_DC	3.22
PH_HD	0.31	PH_HD	0.09	EC_HD	4.48
TEMP_HD	0.29	TEMP_HD	0.08	EC_DC	4.96
DO_HD	0.24	PH_JG	0.07	SEASON_2	5.42
EC_JG	0.19	TEMP_JG	0.06	PH_JG	5.94
PH_DC	0.08	EC_DC	0.05	SEASON_4	6.31
DO_JG	0.07	EC_JG	0.05	DO_JG	8.01
PH_JG	0.07	TEMP_DC	0.02	SEASON_3	8.33
TUR_HD	0.06	DO_JG	0.02	DO_DC	9.17
TEMP_DC	0.06	DO_DC	0.02	PH_HD	16.15
CHLA_JG	0.04	TUR_HD	0.01	TEMP_HD	19.94
DO_DC	0.01	CHLA_JG	0.01	DO_HD	24.56
SEASON_2	0.00	SEASON_2	0.00	PH_DC	25.24
SEASON_3	0.00	SEASON_3	0.00	TEMP_JG	125.43
SEASON_4	0.00	SEASON_4	0.00	TEMP_DC	171.63

Lee et al. (2018) reduced the number of input variables of their ANN model for rainfall prediction from 18 to 5 by using Garson's method and Olden's method. Both methods were used for calculating the relative importance of input variables in an artificial neural network model (Garson, 1991; Olden and Jackson, 2002). The RMSE of the original model using 18 input variables ranged from 85.82 to 87.15 by changing the number of hidden nodes in the ANN structure, where the RMSE of the concise model using five input variables was still maintained similar at 86.84. This exemplary case also shows the importance of selecting a proper method for input variable selection by considering the characteristics of the model algorithm used for prediction.

In general, the input variables were obtained from actual measurements using on-site monitoring tools or manual sample collection and laboratory analysis. However, it was difficult to collect all required input variables in fields, considering labor and related costs of the parameter measurements. The practical implication of the SHAP analysis is that more useful and important input variables can be down-selected by SHAP analysis which showed the best performance among three indices and on-site monitoring and/or measurements can be focused on collecting these selected input variables for model improvement, reducing the cost for water quality monitoring, but improving the Chl-*a* prediction. Therefore, the SHAP analysis was further evaluated to interpret the model performance.

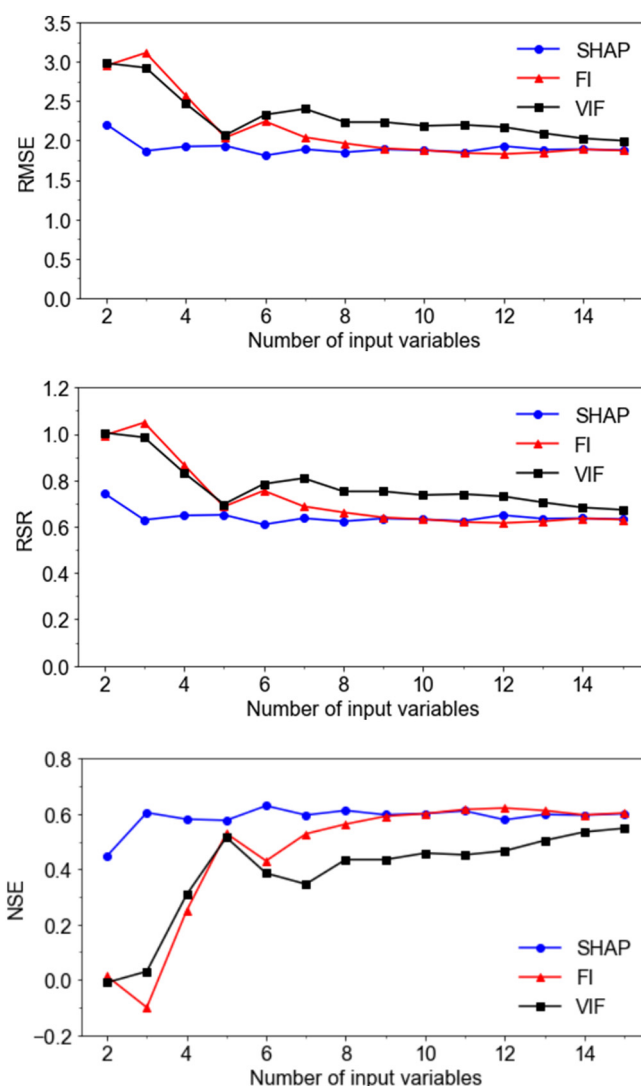


Fig. 3. Model evaluation result with different input variables selection.

3.4. Interpreting model performance

3.4.1. SHAP analysis

A quantitative analysis of the effect of input variables would provide useful information for interpreting and improving the model performance. For example, in a multiple linear regression model, the coefficient of each variable clearly represents the relationship between independent variables and a target-dependent variable. Lee and Lee (2018) developed a multiple linear regression model to predict Chl-*a* concentration in rivers where the coefficients of the five variables (i.e., temperature, pH, DO, BOD, COD) have positive values, representing that the five independent variables are positively correlated with the increase in Chl-*a* concentration. On the other hand, other input variables such as water level have negative coefficients, indicating that the variables are negatively correlated with the change in Chl-*a* concentration.

On the other hand, the effect of input variables on model performance cannot be directly obtained in black box ML models. Thus, a separate algorithm such as SHAP analysis is required to quantify the effect of input variables on the performance of an ML model. The SHAP analysis provided an interpretation of how the model determined the prediction of the dependent variable based on the contribution of input variables to model performance. The SHAP summary plot showed the relative effect of each input variable on model performance (Fig. 4). In Fig. 4, the input variables were sorted by SHAP value so that a variable with a greater effect on model performance was shown at a higher position. The colored dots represented the SHAP value of each sample in the data, whereas the color hue represented the actual value of the observed data from high (red) to low (blue) values. The plot provided an interpretable explanation of how the input variables affected the model prediction. For example, CHLA_DC is located at the top of the figure, which suggests that CHLA_DC had the greatest effect on performance. We can thus estimate that the Chl-*a* concentration of DC station which is located directly upstream of HD station is closely

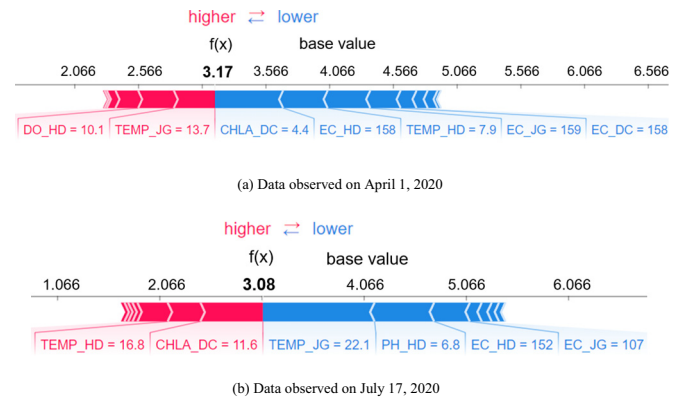


Fig. 5. SHAP force plot.

related to the Chl-*a* concentration of HD station. The red-colored dots located on the right side of Fig. 4 represent positive values of SHAP, while blue-colored dots located on the left side of the figure represent negative values of SHAP. Thus, it could be interpreted that higher CHLA_DC values tended to increase the model prediction of Chl-*a* at HD stations. On the other hand, a higher value of TEMP_JG was represented by red dots located on the left side of the figure, suggesting that a higher value of TEMP_JG tended to decrease the model prediction value.

3.4.2. Interpreting individual events on model results

The SHAP analysis provided a detailed interpretation regarding how the model prediction value was determined for each event. Fig. 5(a) shows one example of the force plot of the data set observed on April 1, 2020, which was used to test the XGB model. The model prediction of Chl-*a* for this observation was 3.17 mg/m³, whereas the actual observed value was 4.4 mg/

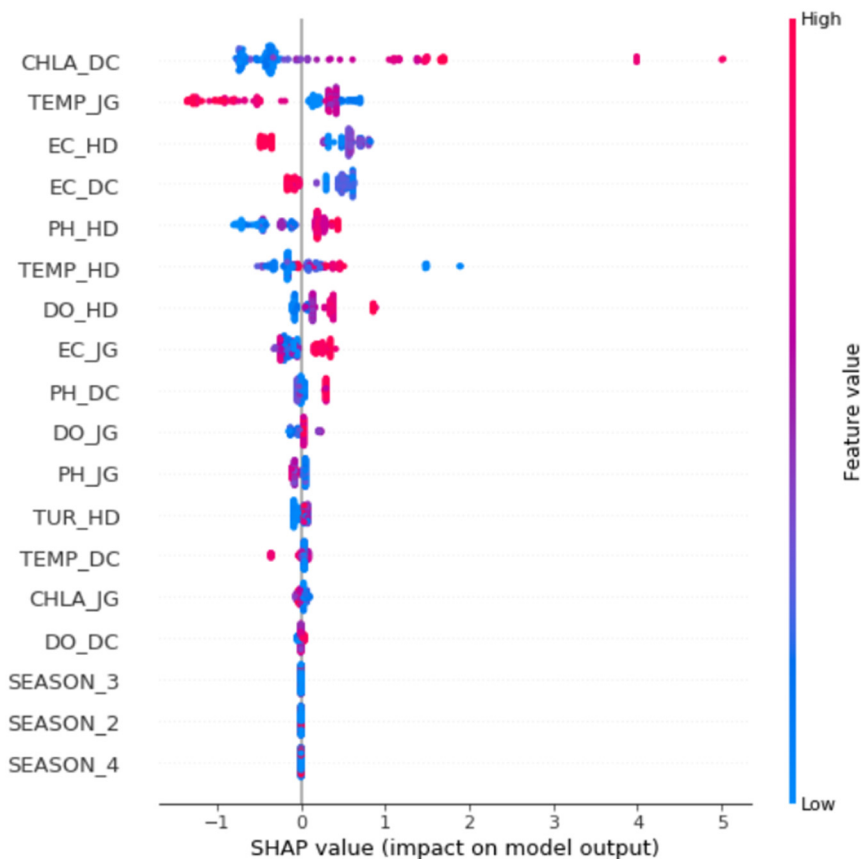
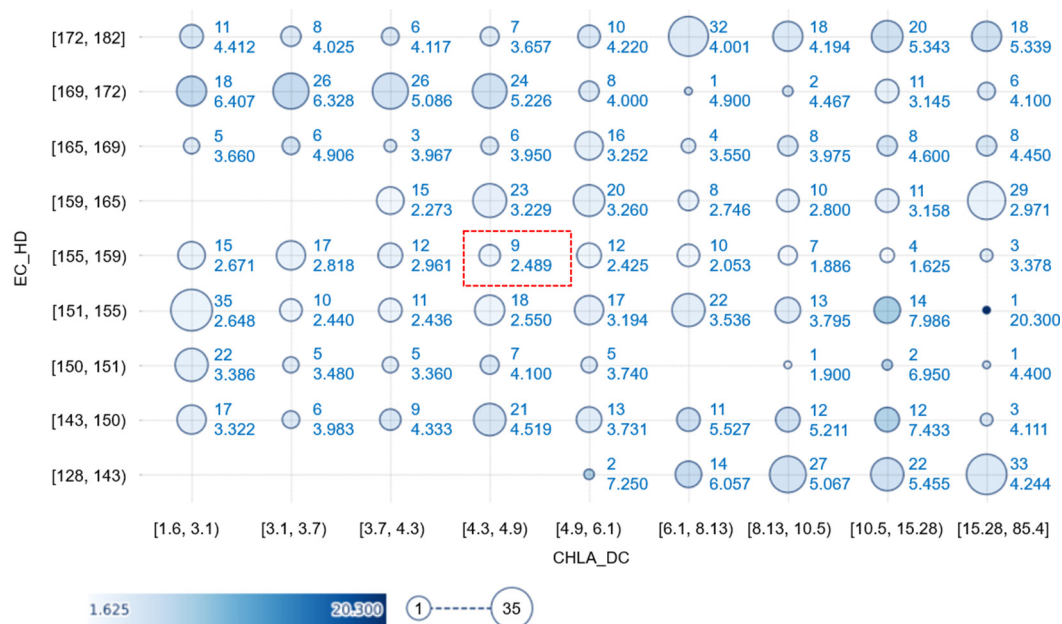
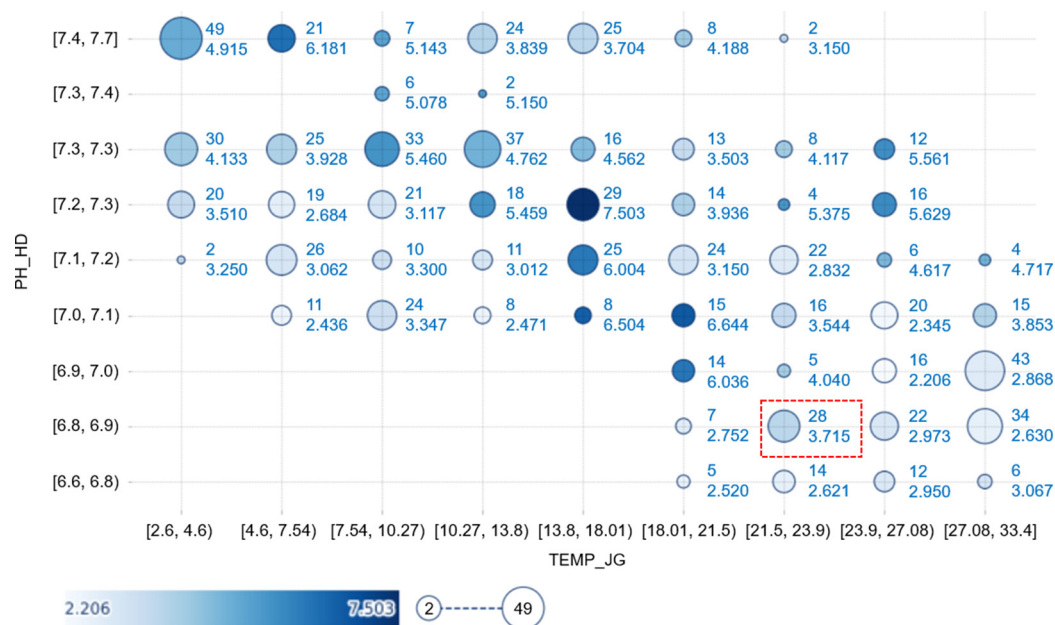


Fig. 4. SHAP summary plot.



(a) Data observed on April 1, 2020



(b) Data observed on July 17, 2020

Fig. 6. Target plot of Chl-*a* at HD station. Note: 1) The color grade represents the average value of CHLA_HD. 2) The size of the circle represents the number of observations within a given range.

m^3 . The base value of Chl-*a* was 4.066 mg/m^3 , which was the average Chl-*a* of all training data sets. The model tended to predict Chl-*a* at the HD station as lower than the average value. The force plot showed that CHLA_DC and EC_HD were two important variables that contributed to this tendency of under-prediction. This SHAP analysis result of the data set observed on April 1, 2020 was clarified by plotting the distribution of average Chl-*a* at

HD station (CHLA_HD) with EC_HD (y-axis) and CHLA_DC (x-axis), which was called the target plot, as shown in Fig. 6(a). The target plot is often used for exploratory data analysis (EDA) to understand the relationship between data and variables through visualization. An open source library called pdpbox was used to produce the target plot of training data in Fig. 6 (PDPBOX, n.d.).

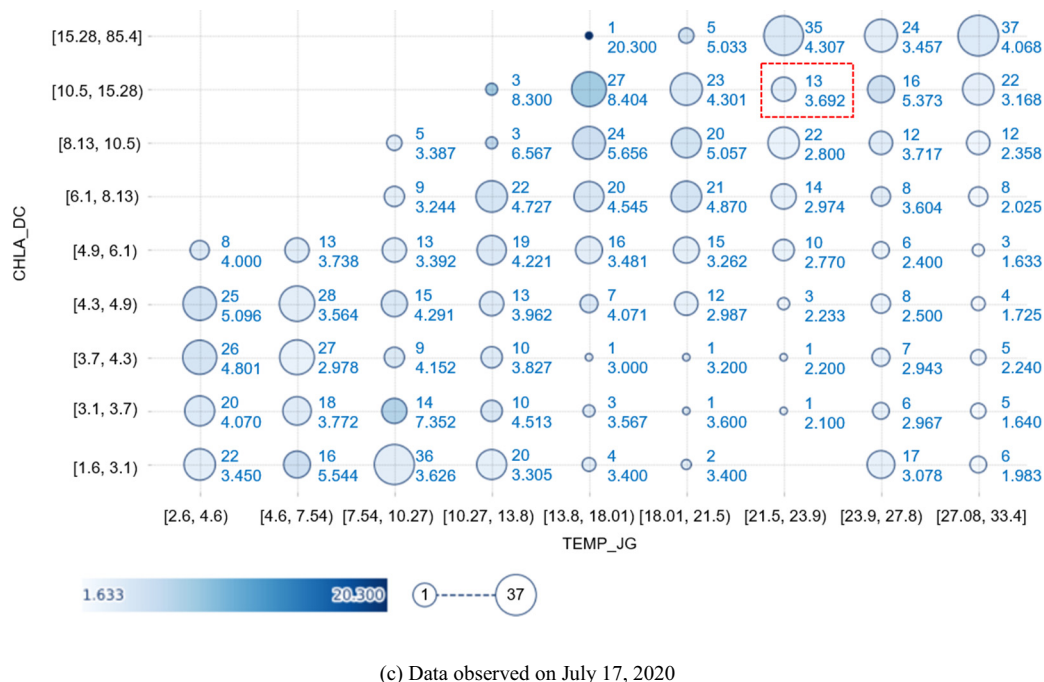


Fig. 6 (continued).

The observed CHLA_DC and EC_HD were 4.4 mg/m³ and 158 μS/cm on April 1, 2020, respectively. This observation fell in the range of CHLA_DC between 4.3 mg/m³ and 4.9 mg/m³, and EC_HD between 155 μS/cm and 159 μS/cm. There were nine observations categorized into this range, and the average CHLA_HD of the nine observations was 2.489 mg/m³, as shown in the red dotted box in Fig. 6(a). This suggested that the observation on April 1, 2020, fell into a lower Chl-*a* range based on values of other input variables, such as CHLA_HD and EC_HD, observed on the same day. Thus, it is estimated that the model tended to predict the Chl-*a* at the HD station (CHLA_HD) as being low value. Fig. 5(b) and 6(b) show another example of a force plot using the data set observed on July 17, 2020, which was used to test the XGB model. The observed Chl-*a* at HD station was 6.2 mg/m³, whereas the model prediction was 3.08 mg/m³ on July 17, 2020. The target plot in Fig. 6(b) shows that the average Chl-*a* at HD (CHLA_HD) was 3.175 mg/m³ in the range categorized by the values of TEMP_JG and PH_HD. The values of TEMP_JG and CHLA_DC also seem to affect the prediction of Chl-*a* at HD to be lower than observed on July 17, 2020. In the range categorized by the values of TEMP_JG and CHLA_DC, the average Chl-*a* at HD (CHLA_HD) was 3.692 mg/m³ as shown in Fig. 6(c), which also provided a possible interpretation about the underprediction of the XGB model. However, it should be noted that the interpretations of these two examples (April 1, 2020 and July 17, 2020) are not the result of the precise analysis, but rather a qualitative interpretation based on quantitative computational results and may vary based on the individual understanding.

SHAP analysis is a popular XAI approach, although it is still in its early stages. Nevertheless, SHAP provided a reasonable and useful interpretation as to how the distribution of input training data affected the results of complex black-box ML models.

3.4.3. Partial dependency plot

PDP provides a more general interpretation of model performance. The abovementioned open-source library pdpbox was also used to plot partial dependency interaction plot (Fig. 7) and partial dependency plot (Fig. 8).

In Fig. 7(a), the x-axis and the y-axis indicate CHLA_DC and EC_HD, respectively. The predicted value of CHLA_HD calculated through the training of the model was visually distinguished by the color bar on the right. In each grid box, the average predicted values (CHLA_HD) of the variables

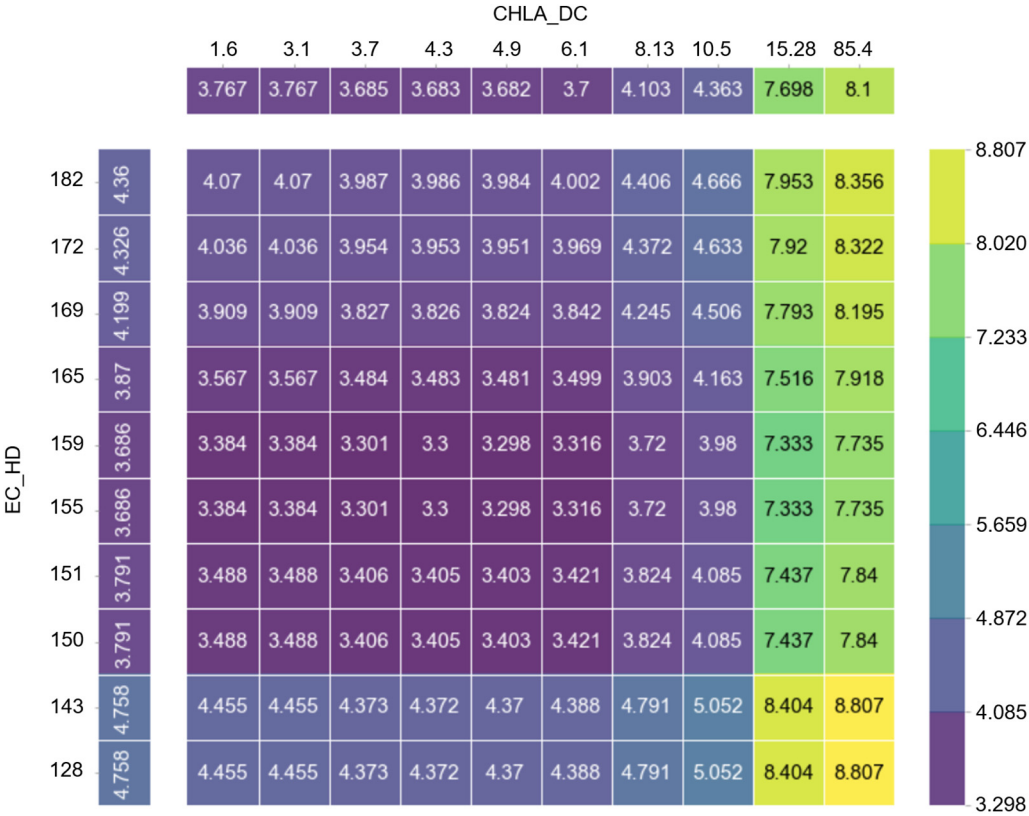
in the corresponding section were entered. Therefore, it is possible to find the values predicted by the model (CHLA_HD) according to the amount of change of the two variables. Fig. 7(a) showed that the larger the CHLA_DC, the larger the predicted value (CHLA_HD), where the effect of EC_HD was relatively less important than that of CHLA_DC. In Fig. 7(b), the x-axis and the y-axis indicate TEMP_JG and PH_HD, respectively. Given the large size of the predicted values in the upper left corner, the smaller the TEMP_JG and the larger the PH_HD, the larger the predicted value (CHLA_HD). Fig. 7(b) also showed that CHLA_HD was smaller on the right side when TEMP_JG was higher than about 20 °C, which corresponded to the previous observation (Fig. 6(b) and (c)).

Fig. 8 shows a representative partial dependence of the predicted value (CHLA_HD) on the increase of TEMP_JG. Therefore, the x-axis indicates the TEMP_JG, and the y-axis denotes the partial dependence of the predicted value (CHLA_HD). Each line represents a partial dependency algorithm for each time; it was observed that as TEMP_JG increased, it had a negative influence on the predicted value (CHLA_HD). Moreover, when TEMP_JG was between approximately 17 and 23, the lines fall sharply, indicating that the range of temperature in JG between 17 and 23 is critical in predicting Chl-*a* in HD as the influence of TEMP_JG on the predicted value (CHLA_HD) in this interval decreased sharply (Fig. 8). This corresponds to the previous observation in Fig. 6(b) and (c).

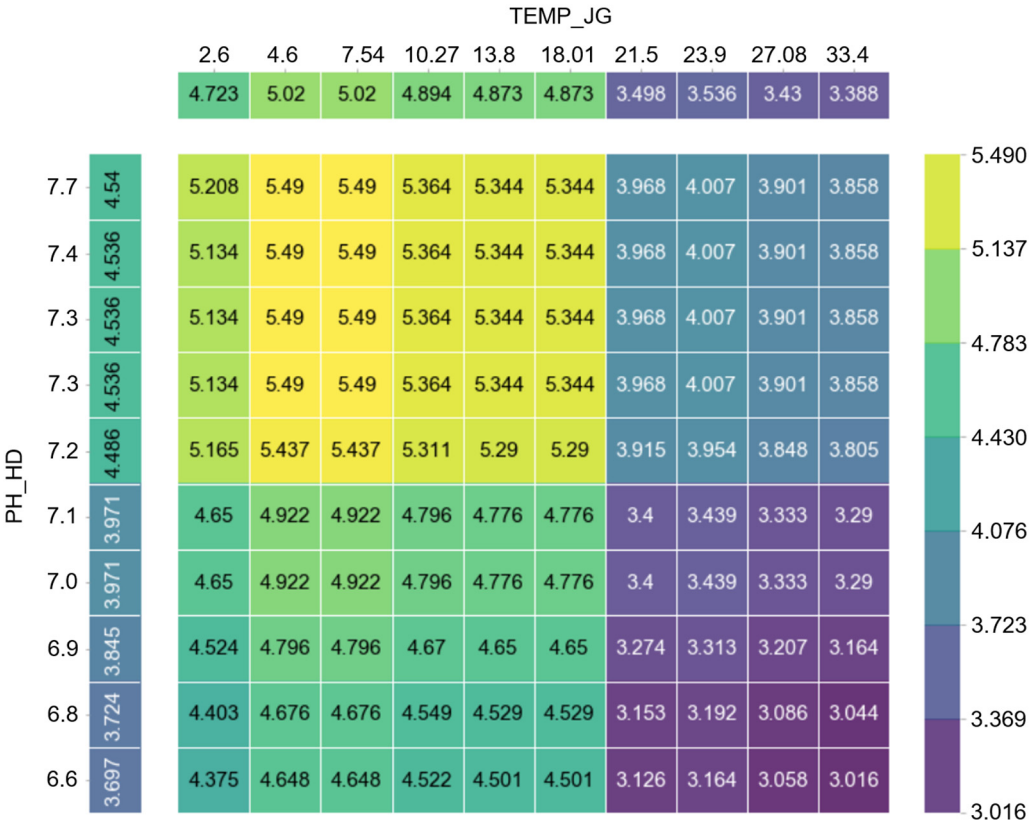
4. Conclusions

Reducing algal bloom in water resources is a major concern in South Korea as well as worldwide, and the quantitative prediction of the status of algal bloom is a primary step in devising a proper management strategy. In recent years, advanced ML models have been increasingly used for algal bloom prediction, and it is necessary to improve the practical applicability of these models. This study provides a scientific approach to improve the practical applicability of an ensemble ML model to predict algal concentration.

First, a novel XAI method was used to reduce the required number of input variables while maintaining a stable ML model performance by analyzing the effect of input variable selection on the model performance. For this purpose, an ensemble ML model called XGB, was developed to



(a)



(b)

Fig. 7. Partial dependency interact plot for CHLA_DC and EC_HD.

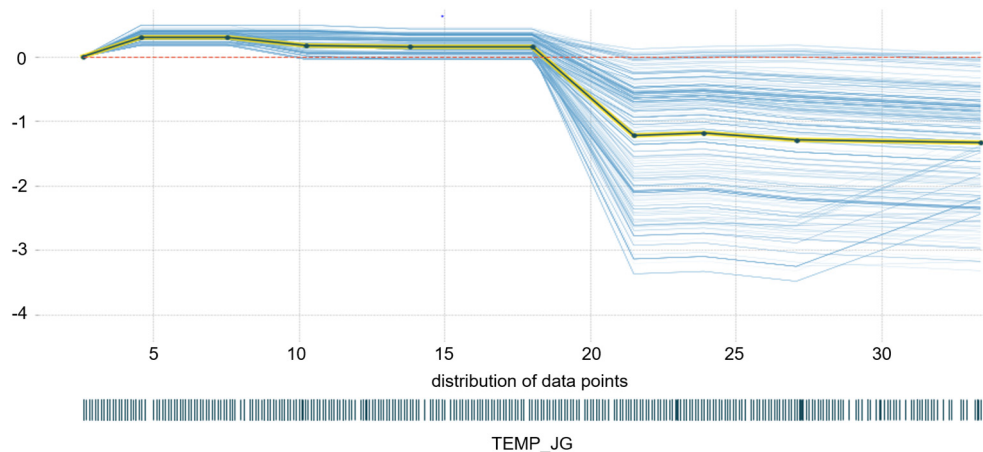


Fig. 8. Partial dependency plot for variable TEMP_JG.

predict Chl-*a* in a stream using 18 independent variables. The model was optimized using a grid search approach, where the values of the evaluation indices were 1.872, 0.630, and 0.603 for RMSE, RSR, and NSE, respectively. Next, the effect of independent input variable selection on model performance was tested further with three indices (i.e., SHAP, FI, and VIF) which have been used to compute the relative importance of each independent variable to determine the model result. Two to 15 independent variables were selectively used in the XGB model. The priority for selecting independent variables was determined using the three indices. The model performance varied based on the number of selected independent variables. The RSR varied from 0.62 to 1.05 and 0.67 to 1.00 with incremental independent variables selected through FI and VIF analysis, respectively. However, the XGB model showed relatively stable performance through the selection of independent variables when SHAP analysis was used, where the RSR ranged from 0.61 to 0.74. This result showed that SHAP analysis, an XAI method, provides useful information that enables a reduction in the required number of independent variables for developing an ML model to reduce the labor and costs involved in field data collection.

Moreover, SHAP analysis of individual observation events combined with the EDA approach provided a reasonable interpretation of how the prediction of the XGB model was determined. By using SHAP values, the effect of each variable on the target value (CHLA_HD) prediction could be interpreted more specifically. Further, through PDA, the independent influences of variables on model prediction could be interpreted and correlations between each variable could be identified in detail. It was possible to conduct a more specific and practical interpretation of the analytical results of the ML model using interpretable learning methods, such as SHAP and PDA. The improved interpretability of the ML model increases the practical applicability of the ML when planning strategies for the proper management of water resources.

ML models have been used in various fields. However, the complicated process inside the model was difficult to understand, and it was considered a drawback and a limitation of black box models. XAI is still in the early stages of development, and this study showed one possible application of the XAI approach to improve ML model performance in predicting water quality.

CRediT authorship contribution statement

Jungsu Park: Conceptualization, Carried out the modeling and data analysis, Investigation, Methodology, Writing - original draft, Writing - review & editing. **Woo Hyoung Lee:** Conceptualization, Investigation, Writing - review & editing. **Keug Tae Kim:** Conceptualization, Investigation, Writing - review & editing. **Cheol Young Park:** Conceptualization, Methodology, Writing - review & editing. **Sanghun Lee:** Carried out the data analysis, Investigation. **Tae-Young Heo:** Conceptualization, Carried

out the modeling and data analysis, Investigation, Methodology, Writing - original draft, Writing - review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported here.

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